SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I Maithreyan.P (192722600089) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: .

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%	2	2	2	2
Method Selection	20%	2	2	2	2
Solution Process	25%	3	2	2	2
Accuracy of Calculation	20%	2	2	2	2
Interpretation of Results	15%	1	1	1	1

Total Marks 37 **/40**

Signature of the Faculty

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ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its interna network. The network administrator has already decided the subnet masks fo each department as follows: HR → /25, Sales → /26, IT → /27, and Admin /28. Using the given subnet masks, apply subnetting techniques to assig suitable subnet addresses for each department. Determine the network addre valid host IP range, and broadcast address for each subnet. Based on t llocation, calculate the number of usable host addresses in each departme

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Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

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Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretatio

SIMATS ENGINEERING

Course: Computer Networks (CSA07)

Assessment Tool 1: Analytical Problem Solving (CO2)

Answers

Question 1: University Labs (192.168.20.0/24)

Given

Block 192.168.20.0/24. Lab A = /26, Lab B = /27, Lab C = /28.

Method

Block size = 2^(32 − prefix). Usable hosts = block size − 2. Subnets are placed one after another, largest first, so each starts on a multiple of its block size. Block sizes: /26 = 64, /27 = 32, /28 = 16.

Solution

Lab	Subnet	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
Lab A	192.168.20.0/26	192.168.20.0	192.168.20.63	192.168.20.1 – 192.168.20.62	62
Lab B	192.168.20.64/27	192.168.20.64	192.168.20.95	192.168.20.65 – 192.168.20.94	30
Lab C	192.168.20.96/28	192.168.20.96	192.168.20.111	192.168.20.97 – 192.168.20.110	14

Interpretation

The three labs use 64 + 32 + 16 = 112 addresses. The range 192.168.20.112 – 192.168.20.255 (144 addresses) is unused and available for future labs.

Question 2: Two Systems with Mask 255.255.255.192

Given

System 1: 192.168.1.130, System 2: 192.168.1.190, both with mask 255.255.255.192 (/26).

Method

Mask 255.255.255.192 is /26, so the block size in the last octet is 256 − 192 = 64. Subnets start at 0, 64, 128, 192. Find the block containing each host value. Usable hosts = 2^6 − 2 = 62.

Solution

Item	System 1 (192.168.1.130)	System 2 (192.168.1.190)
Subnet block	128 – 191	128 – 191
Network address	192.168.1.128	192.168.1.128
Broadcast address	192.168.1.191	192.168.1.191
Valid host range	192.168.1.129 – 192.168.1.190	192.168.1.129 – 192.168.1.190

Item	System 1 (192.168.1.130)	System 2 (192.168.1.190)
Is the IP inside its subnet?	Yes (130 is between 129 and 190)	Yes (190 is the last valid host)
Usable hosts	62	62

Interpretation

Both systems lie in the same subnet, 192.168.1.128/26. They can communicate directly without a router. Note that 192.168.1.190 is the highest assignable host address in that subnet.

Question 3: Departments in 172.16.0.0/16

Given

Block 172.16.0.0/16. HR = /25, Sales = /26, IT = /27, Admin = /28.

Method

Block sizes: /25 = 128, /26 = 64, /27 = 32, /28 = 16. Allocate from the largest to the smallest, starting at 172.16.0.0, so every subnet is correctly aligned.

Solution

Dept.	Subnet	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
HR	172.16.0.0/25	172.16.0.0	172.16.0.127	172.16.0.1 – 172.16.0.126	126
Sales	172.16.0.128/26	172.16.0.128	172.16.0.191	172.16.0.129 – 172.16.0.190	62
IT	172.16.0.192/27	172.16.0.192	172.16.0.223	172.16.0.193 – 172.16.0.222	30
Admin	172.16.0.224/28	172.16.0.224	172.16.0.239	172.16.0.225 – 172.16.0.238	14

Unused range: 172.16.0.240 through 172.16.255.255.

Interpretation

The four departments use 128 + 64 + 32 + 16 = 240 addresses. Of the 65,536 addresses in the /16, 65,296 remain unused, leaving ample room for growth.

Question 4: 172.16.10.0/24 Divided with /27

Given

Network 172.16.10.0/24, subnet mask /27 (255.255.255.224). Hosts to test: 172.16.10.78 and 172.16.10.95.

Method

Borrowed bits = 27 − 24 = 3, so number of subnets = 2^3 = 8. Host bits = 32 − 27 = 5, so usable hosts per subnet = 2^5 − 2 = 30. Block size = 256 − 224 = 32.

Solution

Total subnets: 8. **Usable hosts per subnet:** 30.

Subnet	Network Address	Broadcast Address	Valid Host Range
1	172.16.10.0	172.16.10.31	172.16.10.1 – 172.16.10.30

Subnet	Network Address	Broadcast Address	Valid Host Range
2	172.16.10.32	172.16.10.63	172.16.10.33 – 172.16.10.62
3	172.16.10.64	172.16.10.95	172.16.10.65 – 172.16.10.94
4	172.16.10.96	172.16.10.127	172.16.10.97 – 172.16.10.126
5	172.16.10.128	172.16.10.159	172.16.10.129 – 172.16.10.158
6	172.16.10.160	172.16.10.191	172.16.10.161 – 172.16.10.190
7	172.16.10.192	172.16.10.223	172.16.10.193 – 172.16.10.222
8	172.16.10.224	172.16.10.255	172.16.10.225 – 172.16.10.254

Subnet of 172.16.10.78: 78 lies between 64 and 95, so it belongs to subnet 3. Network address = 172.16.10.64, broadcast address = 172.16.10.95.

Is 172.16.10.95 in the same subnet? Yes, it falls within the range 172.16.10.64 – 172.16.10.95, but it is the broadcast address of that subnet, so it cannot be assigned to a host.

Interpretation

172.16.10.78 is a valid host in subnet 172.16.10.64/27. The address 172.16.10.95 belongs to the same subnet range, but only as its broadcast address, so it is not a usable host address.

Question 5: IPv6 Address Compression and Prefix

Given

2001:0db8:0000:0000:0000:ff00:0042:8329 with prefix length /64.

Method

Rule 1: remove leading zeros in each 16-bit group. Rule 2: replace the longest run of consecutive all-zero groups with "::" (used only once). To expand, restore the zero groups the "::" stands for until there are 8 groups, and pad each group to 4 hex digits.

Solution

Step	Result
After removing leading zeros	2001:db8:0:0:0:ff00:42:8329
Compressed (shortest) form	2001:db8::ff00:42:8329
Expanded (full) form	2001:0db8:0000:0000:0000:ff00:0042:8329
Network prefix (/64)	2001:0db8:0000:0000::/64, written as 2001:db8::/64
Host bits	128 − 64 = 64
Total host addresses	2^64 = 18,446,744,073,709,551,616

Interpretation

The first 64 bits (2001:0db8:0000:0000) identify the network, and the last 64 bits (0000:ff00:0042:8329) form the interface identifier. The subnet holds 2^64 addresses. IPv6 has no broadcast address, so practically all of them can be assigned to hosts.